

Determinants of local and regional communities in intermittent and perennial headwaters of the Bolivian Amazon

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SUMMARY

1. The effect of drying events on aquatic biodiversity is still overlooked in wet Neotropical systems. Yet, the responses of local communities and metacommunities in these biodiversity hotspots may differ from what is reported in other areas.

2. We addressed the effect of drying events on local and regional fish and macroinvertebrate communities in the headwaters of the Chipiriri River basin, in the wet Neotropical piedmont of Bolivia. According to current knowledge in intermittent rivers and ephemeral streams (IRES) ecology, we predicted that intermittent (INT) sites would harbour lower α -diversity, but higher β -diversity, than perennial (PER) sites, due to local, negative effect of drying combined with the existence of multiple recovery stages at the network scale.

3. Although habitat variables were similar between INT and PER sites, local and regional diversity patterns differed. Local invertebrate communities were not different between site types as soon as 4–6 weeks after flow resumption. The proximity of colonist sources and frequent rainfall probably enhanced persistence through dry periods and high resilience. In contrast, fish communities were still poorer at INT than PER sites, indicating they were still in the process of recolonising upstream INT reaches.

4. β -diversity analyses confirmed that invertebrate and fish metacommunities were not at the same recovery stage because (i) β -diversity of invertebrates was best explained by physical and environmental distances at both INT and PER sites, whereas that of fish was explained only by physical distances at INT sites; (ii) fish β -diversity was higher at INT than at PER sites, but invertebrate β -diversity was similar; and (iii) physical distances were correlated with the turnover component of invertebrate β -diversity but with the nestedness component for fish.

5. Exploring regional community patterns in IRES and across biota with different dispersal abilities and modes can advance metacommunity theory and improve our ability to predict local community composition in dynamic ecosystems.

Keywords: dispersal, distance-decay relationships, drying, metacommunity, nestedness, turnover

Introduction

Under pressure of changing climate, and increasing human water abstraction (Gleick, 2003), more streams and rivers are drying (Döll & Schmied, 2012; Acuña

et al., 2014; Datry, Larned & Tockner, 2014b). Ecological changes associated with shifts from perennial to intermittent flow may include the fragmentation of river networks (Jaeger, Olden & Pelland, 2014; Datry *et al.*, 2015), loss of biodiversity (Datry *et al.*, 2014b; Perkin *et al.*,

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2015), alterations in nutrient cycling (Larned *et al.*, 2010), and loss of associated ecosystem services (Boulton, 2014). To be able to accurately predict these changes, one needs to better understand the biodiversity and functioning of naturally intermittent rivers and ephemeral streams (IRES). Following the growing interest in IRES by river scientists (Larned *et al.*, 2010; Datry *et al.*, 2014b; Leigh *et al.*, 2015), knowledge is rapidly accumulating in many parts of the world, including temperate (e.g. Arscott *et al.*, 2010), Mediterranean (e.g. Bonada, Rieradevall & Prat, 2007), alpine (e.g. Rüegg & Robinson, 2004), wet-dry tropical (e.g. Warfe *et al.*, 2013) and arid (e.g. Bogan, Boersma & Lytle, 2013) regions. However, IRES in Neotropical regions, although known as hotspots of biodiversity (Rull, 2011), are still poorly studied (Flecker & Feifarek, 1994; Rincon & Cressa, 2000) probably because of a view that they are rare in very humid areas.

As a master variable in river ecosystems, drying (i.e. the complete loss of surface water) strongly influences biotic communities. With respect to aquatic invertebrates, communities in IRES are generally less speciose and simpler than those of adjacent perennial rivers (e.g. Fritz & Dodds, 2004; Bonada *et al.*, 2007; Grubbs, 2011). Even within single river systems from different climatic zones, sites experiencing drying have, on average, less diverse invertebrate communities than those of nearby perennial sites (e.g. Datry *et al.*, 2014a; Szoëke, Crisman & Thurman, 2015), although they can sometimes be very resilient to drying and recover within a few weeks or months (Rincon & Cressa, 2000; Fritz & Dodds, 2004; Vander Vorste *et al.*, 2015a). Additionally, most studies have found high taxonomic overlap between the communities of perennial and intermittent sites, so that few or no specialists of IRES have been described (Delucchi, 1988; Grubbs, 2011; Datry *et al.*, 2014a). These patterns also hold for fish communities despite their differences in physiological adaptations, dispersal modes and abilities. Within a river system, fish communities in intermittent sites are poorer than their perennial counterparts and consist mostly of generalist species (e.g. Davey & Kelly, 2007; Hoagstrom, Brooks & Davenport, 2011; Perkin *et al.*, 2015). The lower fish diversity at intermittent sites may be the result of slow re-colonisation after drying (i.e. rescue-effect dynamics; Rodriguez, 2010). Moreover, even when sources of colonists are close, fish tend to avoid intermittent sites (Davey & Kelly, 2007; Marshall *et al.*, 2016). Yet, IRES can also be essential for the survival of some fish species, as refuges against predation (Labbe & Fausch, 2000) or spawning sites (Erman & Hawthorne, 1976).

While these patterns have been reported from various climatic and biogeographical areas, they may not hold in wet tropical regions. Several factors may reduce the negative ecological effects of drying in these regions: high productivity and associated biodiversity (Mazzoni & Lobón-Cerviá, 2000; Rull, 2011), abundant rainfall up to 6000 mm yr⁻¹ (Navarro & Maldonado, 2002) facilitating persistence in dry sediments (Stubbington & Datry, 2013), and possible adaptations of Amazonian freshwater organisms to flood pulses, which include tolerance to high temperature and low dissolved oxygen (Bayley, 1995).

Besides uncertainties about the local (α -diversity) effects of drying, little is known about the influence of drying in the spatial organisation of communities at the river network (β -diversity) scale. While spatial community and metacommunity ecology have recently become very popular in community ecology (Leibold *et al.*, 2004; Logue *et al.*, 2011) including in fresh waters (Brown & Swan, 2010; Heino, 2013; Kärnä *et al.*, 2015), they are still poorly explored in IRES (Cañedo-Argüelles *et al.*, 2015; Datry, Bonada & Heino, 2016). It is expected that local (species sorting) rather regional (dispersal) community processes determine the coexistence of species in headwater stream metacommunities, due to their spatial isolation within dendritic networks (Brown & Swan, 2010; Göthe, Angeler & Sandin, 2013). However, this may not apply to IRES where repeated drying events may enhance the importance of dispersal over species sorting (Datry *et al.*, 2016). This is because recolonisation from perennial refuges upon rewetting is generally the dominant pathway supporting community recovery in IRES (Fritz & Dodds, 2004; Bogan *et al.*, 2013; Datry *et al.*, 2014a). Such dynamic may not only lead to different metacommunity patterns in IRES than perennial systems (e.g. strong nestedness), but also promote higher β -diversity at the network scale due to the coexistence of multiple recovery states and/or trajectories depending on the distance to the sources of colonists (Larned *et al.*, 2010; Martin & Ferrer, 2015; Myers *et al.*, 2015). Yet, since these patterns depend on modes and capacity of dispersal (Cañedo-Argüelles *et al.*, 2015; Kärnä *et al.*, 2015), different taxa (e.g. invertebrates and fish) should show contrasting metacommunity patterns at the network scale.

In this study, we addressed the effect of drying events on local and regional fish and macroinvertebrate communities in the headwaters of the Chipiriri River basin, in the wet Neotropical piedmont of Bolivia. We tested whether drying would be as strong a determinant of community structure and composition for both aquatic

invertebrates and fish as observed in temperate, Mediterranean, alpine, wet-dry tropical and arid areas. Following the general community patterns reported in these regions, we predicted that intermittent sites would harbour lower α -diversity, but higher β -diversity, than perennial sites, due to local, negative effects of drying, which result in multiple recovery stages at the network scale.

Methods

Study sites

The Chipiriri River basin (c.100 km² of drainage area) is in the upper Isiboro-Sécure drainage, a headwater of the Amazon River, in the Cochabamba department of Bolivia (Fig. 1a) and comprises eight groundwater-fed headwater tributaries within a highly forested area (Moya, Tomanova & Oberdorff, 2007; Tedesco *et al.*, 2007). At an average elevation of 220 m a.s.l., annual precipitation is 5000–6000 mm, with mean annual air temperature between 24 and 26 °C (Navarro & Maldonado, 2002; Moya *et al.*, 2007). Moreover, these groundwater-fed headwater tributaries display quite homogeneous water chemistry (i.e. average $\pm$ SD ($n = 62$): temperature: 23.6 ± 1.8 °C; pH 7.7 ± 0.3 , specific conductance of 321.5 ± 45.5 $\mu\text{S cm}^{-1}$).

Across these headwater streams, 62 sites were selected (Fig. 1b). Environmental factors were measured and invertebrates were sampled at all sites, including 18 intermittent (INT) and 44 perennial (PER) sites (Fig. 1b). On a subset of 36 sites, including 15 INT sites, fish were also sampled (see below, Fig. 1c). In this area, intermittent sites experience annual drying events lasting from 1 to 4 months, owing to progressive decreases of groundwater levels during the dry period. Drying tends to occur in upstream reaches, although lower reaches can be affected as well (Fig. 1b,c). Apart from drying events, no major natural or anthropogenic pressures occur and flow variability is low, being driven by groundwater levels (Moya *et al.*, 2007; Tedesco *et al.*, 2007).

Environmental variables

At each site, we measured a set of local environmental variables describing the physical habitat. These included mean depth, mean flow velocity, mean wetted width, canopy cover (%) and substratum type. Depth (cm), width (m), flow velocity (cm/s measured 5 cm above the bottom) and substratum type were measured across five stream transects at 3–5 m intervals (depending on

the stream size), with sampling points spaced 1-m apart, and then averaged for each site (Moya *et al.*, 2007; Tedesco *et al.*, 2007). We estimated the average % of substratum in each of six classes (silt and clay; sand; gravel; pebbles; cobbles; boulders). Shading by overhanging vegetation (i.e. percentage of canopy cover) was visually assessed in evenly spaced cross-channel transects.

Invertebrate sampling

Invertebrate sampling took place between June and October 2004, during the dry season. INT sites were sampled after at least 4 weeks of flow had been observed, generally towards the end of the dry season. At each site, five Surber sample units (surface area 0.1 m², mesh size 250 μm) were collected from three to five riffles along the site (c.40 m long reaches) and pooled to form a sample (Moya *et al.*, 2011). Invertebrates were preserved in 5% formalin and further enumerated and identified to the lowest practical level (generally family or genus) in the laboratory using Fernández & Domínguez (2001) and Merritt & Cummins (1996). South American invertebrates are still poorly known, but such coarse resolution was shown to be sufficient to study community patterns across disturbance (e.g. Datry *et al.*, 2014a) or large-scale environmental gradients (e.g. Heino, 2011) and is commonly used for bio-monitoring (e.g. Moya *et al.*, 2011).

Fish sampling

Fish were sampled during the same period at INT and PER sites that had been flowing for at least 4 weeks. At each site electro-fishing was carried out along a length of 40 m to encompass complete sets of geomorphological forms (e.g. riffles and pools). The up- and downstream ends of each site were blocked with nets (1-mm mesh) and two fishing removals were then performed to estimate fish abundance and richness (Tedesco *et al.*, 2007; Ibañez *et al.*, 2009). Fish were counted and identified to the lowest practical level (generally genus or species) using Eigenmann (1918), Pearson (1924) and Géry (1977).

Data analysis

Environmental variables at INT and PER sites. To explore potential differences between INT and PER sites with respect to environmental conditions, a principal component analysis (PCA) was run using the 10 physical habitat variables. A permutational multivariate analysis of

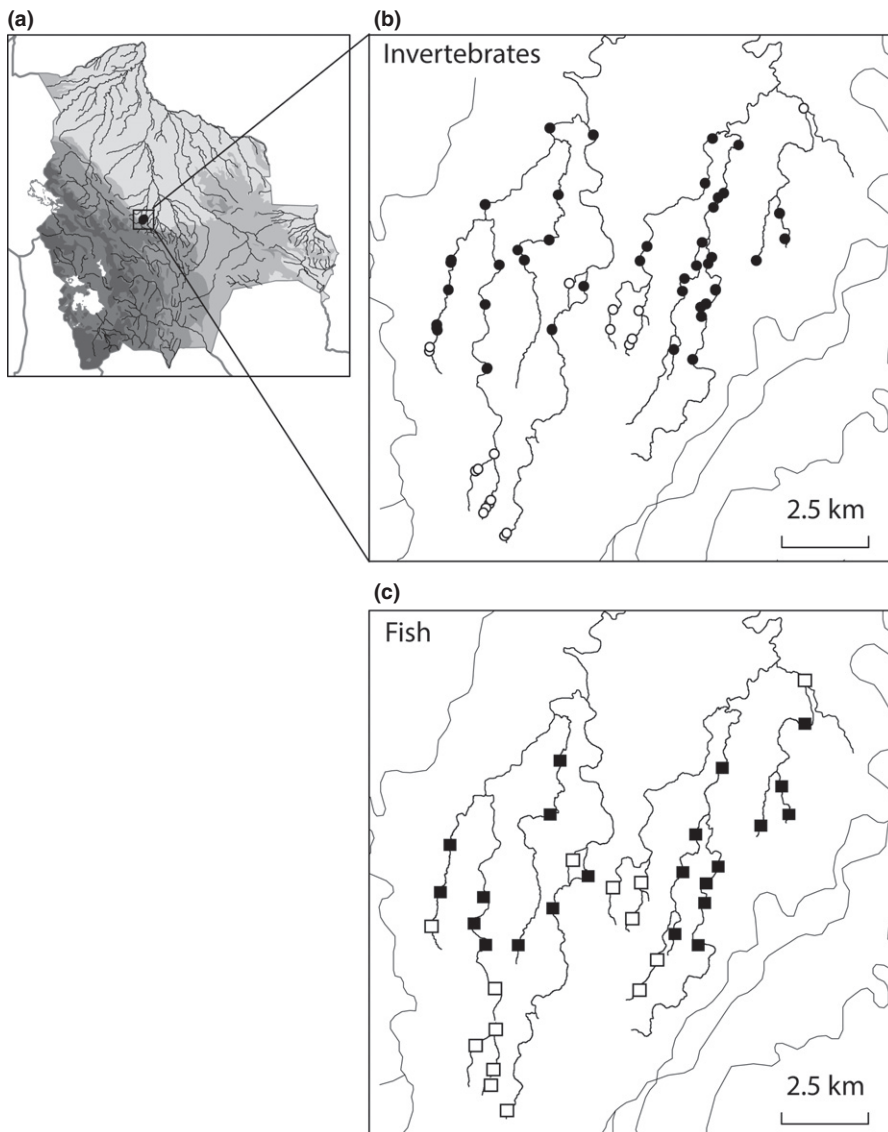


Fig. 1 Sampling sites: (a) the Upper Isiboro-Sécure in Bolivia, (b) invertebrate sampling sites, (c) fish sampling sites. Open symbols: intermittent sites; closed symbols: perennial sites.

variance (ADONIS, Oksanen *et al.*, 2013) was then used to test for differences between INT and PER sites, using Euclidean distances. ADONIS returns a statistic R^2 , which is a measure of separation among groups (0 indicates complete mixing and 1 represents full separation), and a P -value estimated by repeated permutations of the data. Prior to analysis, proportions were $\arcsin(\sqrt{x})$ -transformed to meet normality and homogeneity of variances.

Differences in local community structure and composition between INT and PER sites. For local invertebrate communities, a set of metrics was calculated for each site to test for differences between INT and PER sites (e.g. Bonada *et al.*, 2007; Bogan *et al.*, 2013; Datry *et al.*, 2014a). These included taxonomic richness, density (per m^2), evenness, the proportion of EPT (Ephemeroptera,

Plecoptera, Trichoptera), the proportion of OCH (Odonata, Coleoptera, Hemiptera) and the proportion of DIP (Diptera). Similarly, taxonomic richness, density and evenness of fish communities were calculated for each site. Each fish species was assigned to one of the following categories: omnivorous, insectivorous, detritivorous-herbivorous or piscivorous. Then, one-way analyses of variance (ANOVA) were used to test for differences between INT and PER sites for the each of the above metrics. Density were $\log_{10}(x+1)$ -transformed and proportions were $\arcsin(\sqrt{x})$ -transformed to meet normality and homogeneity of variances.

Compositional changes in invertebrate and fish communities between INT and PER sites were examined using non-metric multidimensional scaling (NMDS). Similarity matrices were calculated using Bray-Curtis distances calculated both on log-transformed densities

and on presence–absence; since both approaches provided similar results, only the results for the log-transformed abundances data are shown, while those using presence–absence are provided in Appendix S1. A permutational multivariate analysis of variance (ADONIS, Oksanen *et al.*, 2013) was then used to test for differences between INT and PER sites. To further assess which taxa were indicative of INT or PER sites, an indicator species analysis (INDVAL) (Dufrene & Legendre, 1997) was conducted.

Differences in metacommunity structure between INT and PER sites. To explore the metacommunity structure between INT and PER site, Mantel tests (Legendre & Legendre, 2012) were used to analyse the relationships between community dissimilarity, environmental variables and physical distances, widely known as the distance-decay relationships framework (DDR, e.g. Nekola & White, 1999; Cañedo-Argüelles *et al.*, 2015; Kärnä *et al.*, 2015). This DDR approach was used for invertebrate and fish communities. Using the 10 environmental variables measured at all sites, a matrix of environmental distances between pairwise combinations of sites was constructed for INT and PER sites, using Euclidean distances, after variable normalisation (mean = 0; SD = 1, Cañedo-Argüelles *et al.*, 2015). Two types of physical distances were produced for each site type (Heino *et al.*, 2012; Göthe *et al.*, 2013; Kärnä *et al.*, 2015). Overland distances were the straight Euclidean distances between pairwise combinations of sites, while watercourse distances were the Euclidean distance between pairwise combinations of sites along the channel network generated via the least-cost path analysis in ArcGIS 9.3 (Environmental Systems Research Institute, Redlands, CA, USA). Community dissimilarity matrices between pairwise combinations of sites were constructed using the Jaccard Podani family index, using the total dissimilarity and its turnover and nestedness components (Legendre, 2014; Datry *et al.*, 2016). Turnover indicates the degree of species replacement by others, from one site to another, whereas nestedness indicates the degree that taxa in species-poor sites are subsets of those in sites with more species and reflects a non-random process explaining the differences in the taxa collected per site due to the loss of taxa (Baselga, 2010; Legendre, 2014). Dissimilarities were based either on log-transformed abundances or presence–absence data; since both approaches provided similar results, only the results for the log-transformed abundances data are shown, while those using presence–absence are provided in Appendix S2. When strong correlations between matrices were detected (Mantel $R > 0.5$ and $P < 0.01$), partial

Mantel tests were used to explore the relationships between community dissimilarity and the matrix of interest, while controlling for the second matrix (Legendre & Legendre, 2012). We did not explore the relationships between fish community dissimilarity and overland distances for sake of ecological meaning.

To test whether β -diversity was higher among INT than PER sites, community dissimilarities between site types were compared for each phyla using the PERMDISP2 routine (Anderson, 2006), a multivariate analogue of Levene's test for homogeneity of variances. Multivariate homogeneity of group variances was calculated using the Jaccard Podani family index, using the total dissimilarity and its turnover and nestedness components. Permutation tests ($n = 9999$) allowed testing for between-site type differences in total, turnover and nestedness components of β -diversity.

For both invertebrate and fish metacommunities, the Chipiriri River network taxonomic richness (γ -richness) was partitioned into site type (INT or PER) and sample richness to assess the respective contribution of α -sample and β -site type richness to γ -richness (Crist *et al.*, 2003). A complete randomisation procedure was used to generate 10 000 random distributions of invertebrate and fish species among samples at all hierarchical levels (i.e. sample, site type) to generate the null distribution of each α - and β -estimate for each level (Crist *et al.*, 2003). Level-specific estimates were compared with appropriate null distributions and used to test the null hypothesis that the observed α - and β -diversity were generated by a random distribution of species among samples. Statistical significance was assessed by the proportion of null values greater or smaller than the actual estimate (Crist *et al.*, 2003). Separate diversity partitioning analysis was applied (i) to the entire community; (ii) to the community of taxa representing $>5\%$ of individuals and (iii) to the community of taxa representing $<0.1\%$ of individuals, to evaluate whether common and rare taxa contributed to α - and β -diversity.

All statistical analyses were carried out using R software (R Core Team, 2013) and functions in the package Vegan (Oksanen *et al.*, 2013) and Betapart (Baselga & Orme, 2012). For all tests, $P < 0.05$ indicated statistical significance.

Results

Environmental conditions at INT and PER sites

The first two axes of the PCA explained 46.7% of the total variance in the environmental variables across all

sites (Fig. 2). The first axis (25.2%) represented a streambed sediment gradient, being positively correlated with % pebble, % cobble and % gravel and negatively with % clay and water depth (Fig. 2a). The second axis (21.5%) represented a longitudinal gradient, being negatively correlated with stream width, velocity, water depth and % riparian cover (Fig. 2a). The distribution of sites in this multivariate space indicated that INT sites were predominantly located in headwaters (Fig. 2b, Table 1). However, the ADONIS showed that environmental variables did not differ between INT and PER sites (ADONIS, $P = 0.1107$, Table 1).

Local communities: differences between perennial and intermittent sites

We identified 33 933 invertebrates belonging to 45 taxa (Appendix S3). The most abundant taxa were Leptophlebiidae (Ephemeroptera; relative abundance = 22.6%), Chironomidae (Diptera, 22.2%) and Elmidae (Coleoptera, 13.5%). Twenty-six taxa had relative abundance < 1% (Appendix S3). INT and PER sites had similar density (ANOVA, $P = 0.4852$, Fig. 3) and taxonomic richness (ANOVA, $P = 0.1519$, Fig. 3). Similarly, there was no difference in taxa evenness between INT and PER sites (ANOVA, $P = 0.4159$, Fig. 3). Last, the proportion of EPT, OCH and DIPT were not different between INT and PER sites (ANOVA, $P = 0.8306$, 0.1159, 0.426, respectively, Fig. 3).

There was no difference in local invertebrate community composition between INT and PER sites when using log-transformed density (ADONIS, $P = 0.3614$, Fig. 4). However, when based on presence-absence, community similarities differed slightly between INT and PER sites (ADONIS, $P = 0.023$, Appendix S1). Indicator species analyses indicated that Libellulidae (Odonata) and Palaemonidae (Decapoda: Crustacea) were indicative of INT sites ($P = 0.008$ and 0.032 respectively), whereas Odontoceridae (Trichoptera) and Psephenidae (Coleoptera), Naucoridae (Hemiptera) and Corydalidae (Megaloptera) were indicative of PER sites ($P = 0.002$, 0.003, 0.031 and 0.042 respectively).

A total of 17 263 fish individuals were identified, belonging to 69 taxa (Appendix S4). The three most abundant taxa were characids: *Astyanax abramis* (relative abundance = 22.3%), *Odontostilbe dierythrura* (9.6%) and *Hemigrammus cf lunatus* (9.1%). Forty-eight taxa had relative abundance < 1% (Appendix S4). Fish density and taxonomic richness were lower in INT than PER sites (ANOVA, $P = 0.0035$ and $P < 0.001$, respectively, Fig. 5). There was no difference in taxon evenness between INT

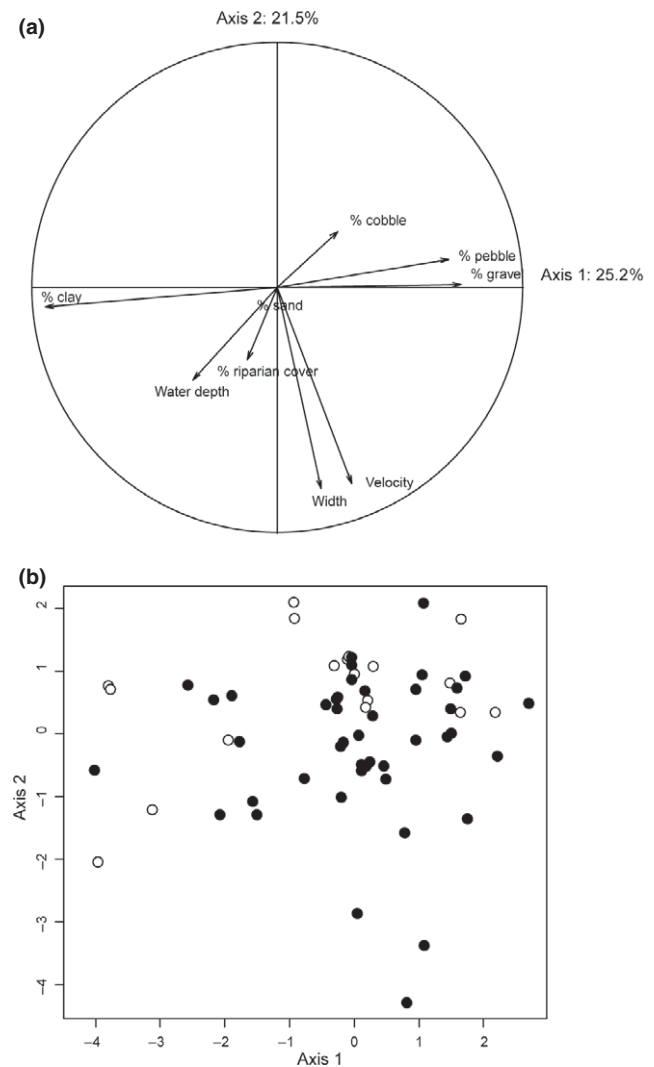


Fig. 2 Principal Component Analysis (PCA) of site environmental variables: (a) correlation circle, (b) distribution of sites ($n = 62$) along axes 1 and 2 (46.7% of the total variance). White circles: INT sites; black circles = PER sites.

and PER sites (ANOVA, $P = 0.8822$, Fig. 5). The proportion of insectivorous fish was slightly higher in INT than in PER sites, as was that of piscivorous fish (ANOVA, $P = 0.042$ and 0.049 , respectively, Fig. 5). The proportions of detritivorous-herbivorous and omnivorous fish did not differ between stream types (Fig. 5).

Local fish community composition differed between INT and PER sites both when using log-transformed density (ADONIS, $P < 0.001$, Fig. 6) and presence-absence data (ADONIS, $P < 0.001$, Appendix S1). Indicator species analyses indicated that *Astyanax abramis* ($P < 0.001$), *Leporinus striatus* ($P < 0.001$), *Characidium spA* ($P = 0.004$), *Astyanacinus multidentis* ($P = 0.003$), *Satanoperca jurupari* ($P = 0.007$), *Parodon cf buckleyi*

Table 1 Environmental variables at PER and INT sites

	PER				INT			
	Min	Max	Mean	SD	Min	Max	Mean	SD
Depth (cm)	8.62	65.00	23.12	12.56	9.40	50.81	22.62	13.97
Flow velocity (m s^{-1})	0.00	0.98	0.28	0.25	0	0.48	0.16	0.14
Wetted width (m)	2.83	19.00	6.74	3.41	1.63	6.76	4.75	1.51
% riparian cover	68.00	95.00	84.00	5.00	54.00	95.00	85.00	10.00
% silt/clay	0.61	83.30	31.95	14.63	0.00	99.06	40.48	38.42
% sand	2.60	44.74	13.66	8.43	0.00	34.56	8.77	11.92
% gravel	0	50.70	21.08	7.54	0.00	53.46	19.33	19.40
% pebbles	1.10	75.00	30.61	14.23	0.00	68.00	15.78	20.64
% cobbles	0.00	17.28	2.69	2.58	0.00	29.11	5.64	8.59
% blocks	0.00	2.00	0.10	0.40	0.00	0.10	0.05	0.1

Min, minimum; Max, maximum; Mean, mean; SD, standard deviation.

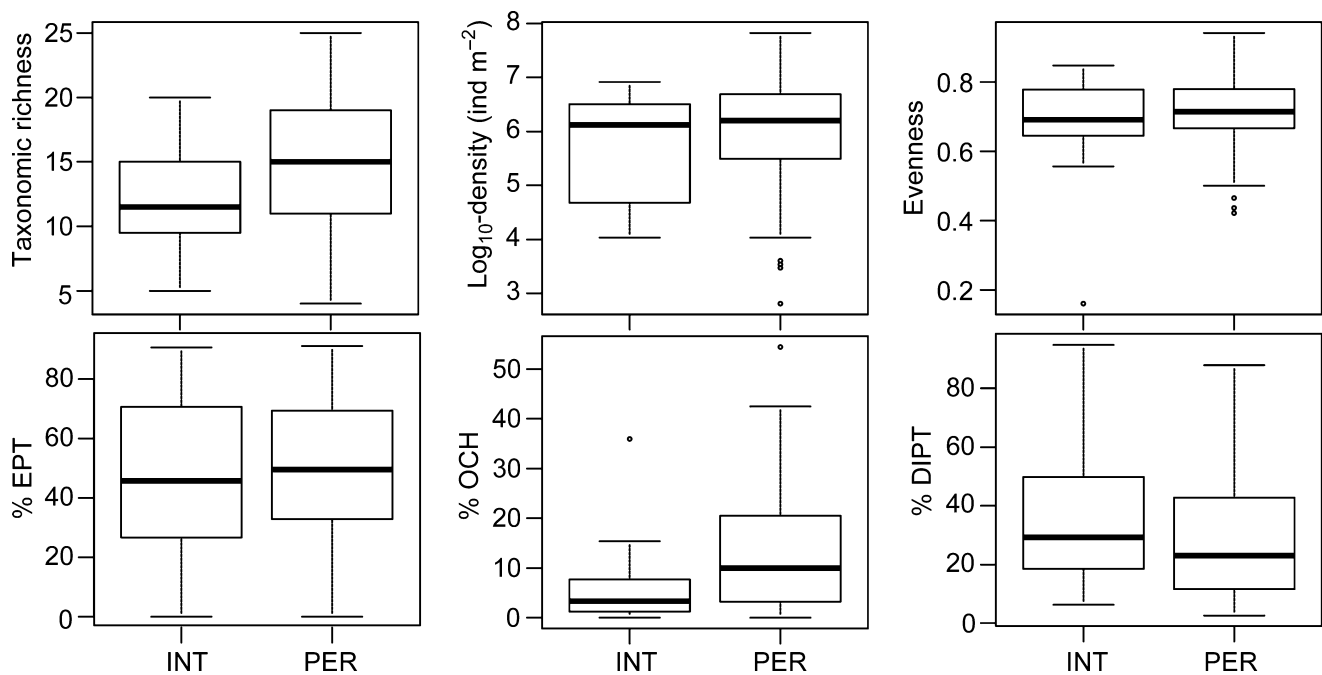


Fig. 3 Box plots of invertebrate metrics (see data analysis for definitions) for INT and PER sites. The horizontal lines represent the first, second (median) and third quartiles, and open circles are outliers.

($P = 0.009$), *Hemibrycon* sp. ($P = 0.017$), *Steindacnerina dobula* ($P = 0.017$) and *Astianax lineatus* ($P = 0.043$) were indicative of PER sites. In contrast, only *Hypostomus gr. cochliodon* was indicative of INT sites ($P = 0.040$). This species was found exclusively at 3 INT sites.

Differences in metacommunity structure between INT and PER sites

Invertebrate community dissimilarity at INT sites was explained by environmental, overland and water course distances (Mantel $R = 0.40$, 0.39 and 0.22 , respectively,

Table 2a) and this held after controlling for the correlations among distance matrices (Table 2b,c). In particular, the turnover component of community dissimilarity was highly correlated with overland distances, while the nestedness component was not related to either overland or watercourse distances (Table 2a,b). In contrast, invertebrate community dissimilarities at PER sites were explained only by environmental distances (Mantel $R = 0.20$) and this held after controlling for overland and watercourse distances (partial Mantel $R = 0.19$ and 0.18 respectively). After controlling for environmental distances (Table 2b), invertebrate community dissimilarities

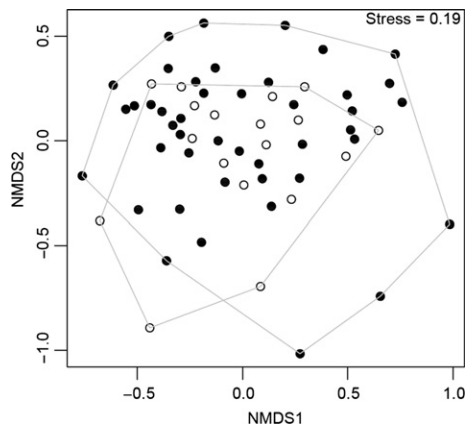


Fig. 4 NMDS ordinations of sites based on local invertebrate communities using log-transformed abundance. Grey circles: INT sites ($n = 18$); black circles: PER sites ($n = 44$).

at PER sites were not correlated with either overland or watercourse distances. In general, watercourse distances explained less invertebrate community dissimilarity than overland ones (Table 2a,b).

Fish community dissimilarity at INT sites was explained by watercourse distances (Mantel $R = 0.32$, Table 2a), especially after controlling for environmental distances (partial Mantel $R = 0.31$, Table 2b), but not by environmental distances (Table 2a,c). Only the nestedness component of fish community dissimilarity at INT sites was correlated with physical distances (Table 2b). In contrast, fish community dissimilarity at PER sites was explained by environmental distances (partial Mantel $R = 0.26$ after controlling for watercourse distances), but not by watercourse distances (Table 2b,c).

Total β -diversity of invertebrate communities was not different between INT and PER sites (Permdisp, $P = 0.2164$, Fig. 7a), but its turnover component was higher at INT than PER sites (Permdisp, $P = 0.021$, Fig. 7b). The nestedness component did not differ between site types (Permdisp, $P = 0.2249$, Fig. 7c). In contrast, total β -diversity of fish communities was higher at INT than PER sites (Permdisp, $P = 0.002$, Fig. 7d), but turnover or nestedness components were not different between the site types (Permdisp, $P = 0.224$ and 0.301 , respectively, Fig. 7e,f).

When considering all invertebrate taxa, total richness was predominantly explained by α -sample richness (62.4%, Table 3a). Additive partitioning indicated consistent and significantly higher α -sample richness than expected by chance and a lower β -sample and β -site type richness than expected by chance (Table 3a). Total richness of rare taxa was predominantly explained by β -sample richness (59.7%) although this was not different than expected by chance (Table 3a). Most of the richness in common taxa was explained by α -sample richness and this was higher than expected by chance (95.0%, Table 3a). In contrast, β -sample richness was lower than expected by chance and contributed only 1.0% to the total richness of common taxa, while the contribution of β -site type richness was not greater than expected by chance (Table 3a).

When considering all fish taxa, total richness was largely explained by α -sample richness (63.7%, Table 3b). Additive partitioning indicated consistent and significantly higher α -sample richness than expected by chance and a lower β -sample and β -site type richness than

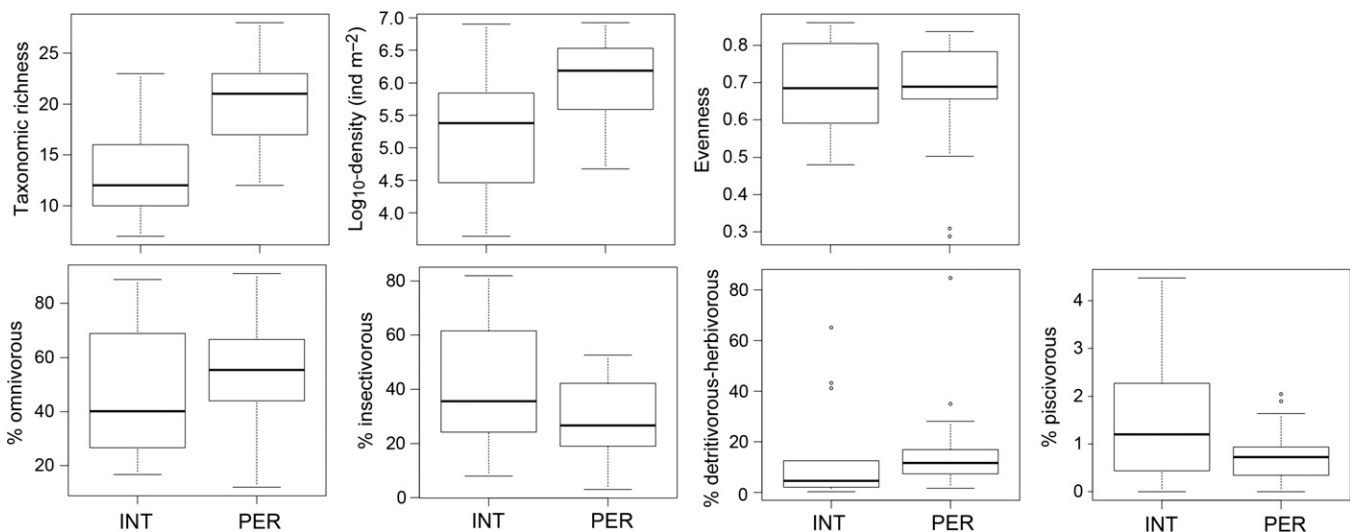


Fig. 5 Box plots of fish metrics (see data analysis for definitions) for INT and PER sites. The horizontal lines represent, respectively, the first, second (median) and third quartiles, and open circles are outliers.

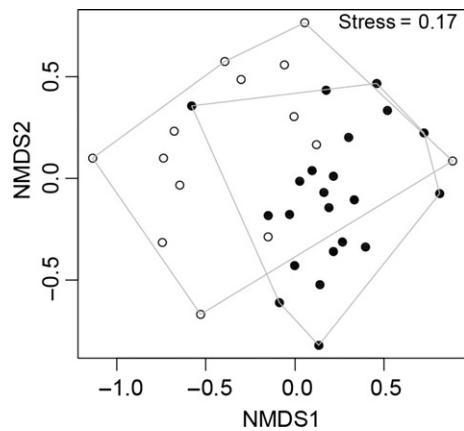


Fig. 6 NMDS ordinations of sites based on local fish communities using log-transformed abundance. Grey circles: INT sites ($n = 14$); black circles: PER sites ($n = 22$).

expected by chance (Table 3b). Total richness in rare species was predominantly explained by β -sample richness (50.1%) although this was not different than expected by chance (Table 3b). Most of the richness in common species was explained by α -sample richness and this was higher than expected by chance (99.5%, Table 3b). In contrast, β -sample richness was lower than expected by chance and contributed only 0.5% to the total richness of common taxa, while the contribution of β -site type richness was not greater than expected by chance (Table 3b).

Discussion

In this study, we compared aquatic invertebrate and fish communities between INT and PER sites in headwater streams in the rainy Neotropical piedmont in Bolivia. At the local scale, drying had little effect on invertebrate communities, but greater effects on fish. At the regional scale, invertebrate and fish metacommunities showed contrasting organisation, probably reflecting contrasting colonisation dynamics. Still, the presence of INT sites did not increase β -diversity at the regional scale. Overall, our results suggest that the effect of drying events on IRES communities may not be as severe as those in other climatic areas. This has strong implications for predicting the future effects of global change in this area, now prone to increased human development.

Invertebrate and fish communities are differently affected by drying

In the Chipiriri headwaters, environmental characteristics of INT and PER sites were similar under flowing

conditions, despite INT sites being predominantly located in the upstream sections. The relatively short distance between sites (average overland distance among sites $\pm$ SD = 5.1 ± 2.7 km) probably explains such environmental similarity, which in turn allowed us to focus on flow regime differences between site types. INT and PER sites had similar invertebrate communities. Additionally, and as reported in many IRES (e.g. Delucchi, 1988; Bonada *et al.*, 2007; Datry *et al.*, 2014a), including a tropical one (Rincon & Cressa, 2000); broad taxonomic overlap was found between INT and PER sites, and only two and four taxa were indicative of INT and PER sites respectively. The drying events, which can last between 1 and 4 months in this area, were not as important a determinant of invertebrate communities as in arid (e.g. Bogan *et al.*, 2013), alpine (e.g. Rugg & Robinson 2004), Mediterranean (e.g. Bonada *et al.*, 2007), tropical (e.g. Warfe *et al.* 2013), or temperate (e.g. Arscott *et al.*, 2010) regions. For example, in a cross-continental study of 14 different rivers encompassing the above climate types, Datry *et al.* (2014a) found that invertebrate taxonomic richness, density and % EPT were lower at INT than PER sites. This was true at different levels of taxonomic resolution, including the family level, although we cannot exclude the possibility that this coarse resolution could have partly masked subtle species-level differences between sites.

The lack of differences between INT and PER in local invertebrate communities could be related to fast and complete recolonisation of intermittent headwaters from nearby colonist sources, and rapid dispersal of organisms. In others systems, high community resilience was explained by the proximity of colonist sources (Fritz & Dodds, 2004; Chester & Robson, 2011; Vander Vorste *et al.*, 2015a). Another possible explanation for the lack of differences between INT and PER communities is that desiccation-resistant forms survived under dry conditions (Stubbington & Datry, 2013). High humidity and frequent rainfall in the region (Navarro & Maldonado, 2002) may have enhanced the significance of desiccation-resistance above what occurs in temperate, desert and Mediterranean areas (e.g. Boulton *et al.*, 1992; Bilton, Freeland & Okamura, 2001; Stubbington & Datry, 2013). As many as 12 invertebrate families have been collected from dry streambed sediments in this region (Datry *et al.*, unpubl. data). Lastly, most sites in the Chipiriri River basin had coarse substrata and shallow ground water (<1 m) which could have made the hyporheic zone a source of colonists (Stubbington, 2012; Vander Vorste, Malard & Datry, 2015b). In contrast, it is very unlikely that mass-effect events had occurred (Leibold

Table 2 Results of Mantel tests and their partial derivatives for abundance data based on Jaccard (Podani family) dissimilarity matrices. (a) correlations between community dissimilarities, environmental (ENV) and physical (LAND = overland, CHANNEL = watercourse) distances; (b) correlations between community dissimilarities and physical distances (LAND = overland, CHANNEL = watercourse) controlling for environmental distances (ENV); (c) correlations between community dissimilarities and environmental distances (ENV) controlling for physical distances (LAND = overland, CHANNEL = watercourse)

(a)				
Site type	Organism	ENV	LAND	RIVER
INT	Invertebrates			
	Total dissimilarity	0.40***	0.39 ***	0.22*
	Turnover	0.19 (NS)	0.46 ***	0.14 (NS)
	Nestedness	0.20 (NS)	−0.18 (NS)	−0.06 (NS)
	Fish			
	Total dissimilarity	−0.07 (NS)		0.32***
	Turnover	0.13 (NS)		0.03 (NS)
	Nestedness	−0.14 (NS)		0.06 (NS)
PER	Invertebrates			
	Total dissimilarity	0.20 **	−0.03 (NS)	0.04 (NS)
	Turnover	−0.02 (NS)	0.05 (NS)	0.05 (NS)
	Nestedness	0.13 *	−0.03 (NS)	−0.01 (NS)
	Fish			
	Total dissimilarity	0.29**		0.18*
	Turnover	0.06 (NS)		0.01 (NS)
	Nestedness	0.12 (NS)		0.06 (NS)
(b)				
Site type	Organism	LAND		RIVER
INT	Invertebrates			
	Total dissimilarity	0.24 *		0.18*
	Turnover	0.42***		0.07 (NS)
	Nestedness	−0.19 (NS)		−0.05 (NS)
	Fish			
	Total dissimilarity			0.31***
	Turnover			0.01 (NS)
	Nestedness			0.21**
PER	Invertebrates			
	Total dissimilarity	−0.01 (NS)		0.03 (NS)
	Turnover	0.04 (NS)		0.05 (NS)
	Nestedness	−0.04 (NS)		−0.02 (NS)
	Fish			
	Total dissimilarity			0.11 (NS)
	Turnover			−0.05(NS)
	Nestedness			0.03 (NS)
(c)				
Site type	Organism	LAND		RIVER
INT	Invertebrates			
	Total dissimilarity	0.28**		0.39**
	Turnover	−0.05 (NS)		0.17 (NS)
	Nestedness	0.27**		0.19 (NS)
	Fish			
	Total dissimilarity			−0.04 (NS)
	Turnover			0.11 (NS)

Table 2 (Continued)

PER	Nestedness		0.06 (NS)
	Invertebrates		
	Total dissimilarity	0.19 **	0.18**
	Turnover	−0.04 (NS)	−0.05 (NS)
	Nestedness	0.11 (NS)	0.12 (NS)
	Fish		
	Total dissimilarity		0.26**
	Turnover		−0.06 (NS)
	Nestedness		0.14 (NS)

Values are Mantel or partial Mantel *R* and associated *P*-value: NS $P > 0.05$, * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$.

et al., 2004) as the density of invertebrates was not overly high compared to those of systems in other regions (e.g. Fritz & Dodds, 2004; Bogan *et al.*, 2013; Vander Vorste *et al.*, 2015a), suggesting productivity was not especially high. Further work will explore these diverse sources of resilience.

In contrast to invertebrates, fish communities of INT and PER sites were markedly different, both in terms of structure and composition, paralleling findings from temperate (e.g. Davey & Kelly, 2007), Mediterranean (e.g. Boughton *et al.*, 2009; but see Magalhaes *et al.*, 2007) and desert IRES (e.g. Balcombe & Arthington, 2009). In spite of the proximity of sites (average watercourse distance among all sites = 27.9 ± 17.5 km), INT sites had consistently fewer species and lower density of fish, and different community composition, than PER sites, despite no difference in habitat. Importantly, no desiccation-resistant fish are known from the region, and this could partly explain the contrasts between invertebrate and fish. Therefore, INT-PER differences are likely due to the predominantly upstream position of INT sites restricting their fast recolonisation. Directional stream flow makes the position of INT sites within the river longitudinal gradient important in recolonisation (Davey & Kelly, 2007; Datry *et al.*, 2014a). Upstream location may not restrict the recolonisation of most invertebrates that have either passive or active aerial colonisation (Bilton *et al.*, 2001), whereas watercourse distances, 6-fold higher than overland ones, was potentially a strong limiting factor for fish. Consequently, it may take several months until fish communities from both site types become similar, if at all. Indeed, differences between INT and PER rivers/sites in fish (e.g. Davey & Kelly, 2007; Balcombe & Arthington, 2009; but see Magalhaes *et al.*, 2007) and invertebrate (e.g. Arscott *et al.*, 2010; Datry, 2012) communities can persist over entire flow periods. In the Chipiriri River basin, this pattern may occur due to lower productivity at INT sites due to reduced leaf litter decomposition during the periods of

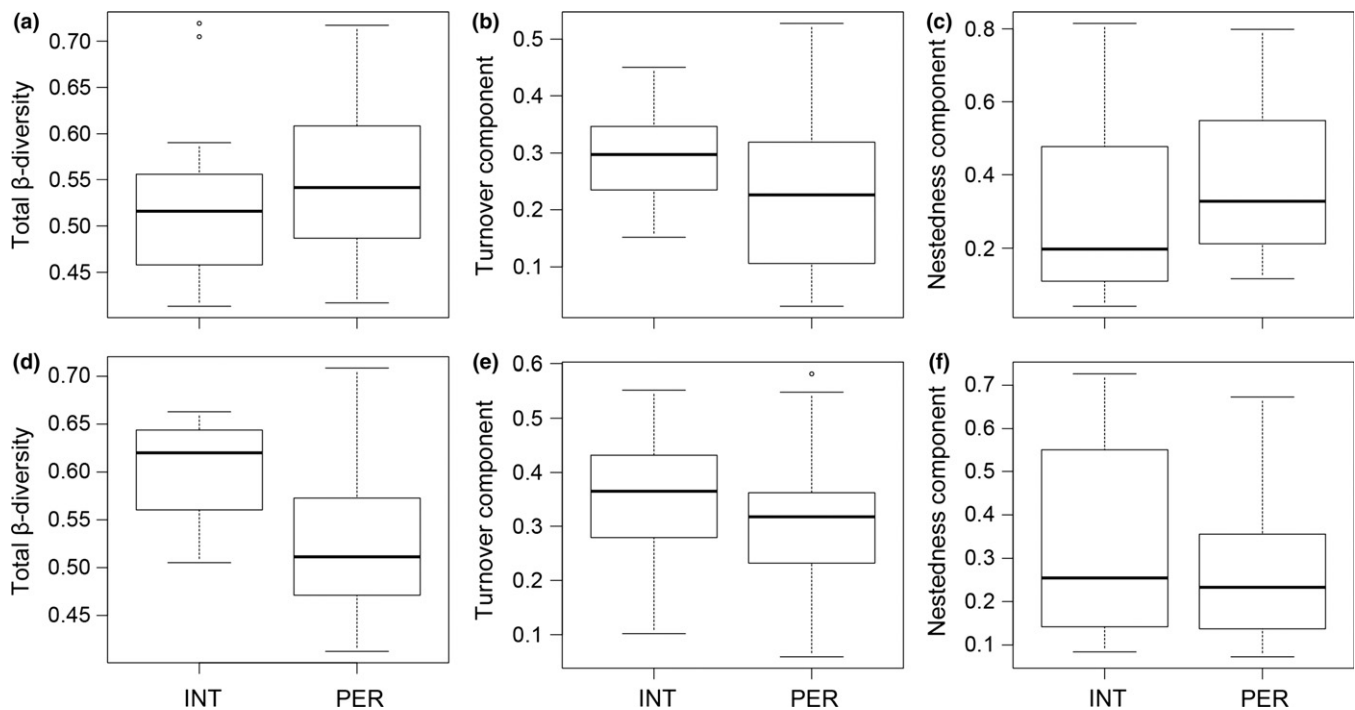


Fig. 7 Differences between INT and PER site in total β -diversity, turnover and nestedness components for invertebrate (a–c) and fish (d–f) communities.

drying. Allochthonous organic matter entering the system throughout the year is the major basal energy source for these tropical forested streams, far exceeding autochthonous primary production, which is light-limited by riparian cover (Tedesco *et al.*, 2007). The decomposition of this litter is severely reduced under lentic or dry conditions (Sangiorgio, Fonnesu & Mancinelli, 2007; Corti *et al.*, 2011).

Both invertebrate and fish metacommunity patterns are driven by dispersal at intermittent sites

Unlike local community structure and composition, invertebrate and fish metacommunities of INT and PER sites showed similar patterns, which indicates drying could increase the role of dispersal in shaping metacommunities. Importantly, apart from drying, there is no barrier to dispersal in the Chipiriri headwaters: the area is flat, the flow regime is stable and human presence is minimal. Community dissimilarity for both invertebrates and fish across INT sites was highly correlated with physical distances (overland and water course distances respectively), whereas dissimilarity across PER sites was correlated with environmental distances only. These patterns confirm that, at the regional scale, INT communities were recovering when they were sampled, with

colonists from PER sites gradually reaching the upstream INT sites (Cañedo-Argüelles *et al.*, 2015). Yet, the recovery stages probably differed between groups. For invertebrates, the turnover component of β -diversity was correlated with physical distances, whereas for fish, it was the nestedness component. This confirms the local community patterns discussed above and indicates that fish were still colonising INT sites at the sampling period, while invertebrates had already reached most INT sites. Local invertebrate communities from INT sites exhibited no differences in structure and composition from those of PER sites and within INT sites, there was a dissimilarity turnover vs. physical distance relationships. In contrast, fish communities differed between INT and PER sites and within INT sites, there was dissimilarity nestedness vs. physical distance relationships, suggesting ongoing recolonisation. In contrast, community dissimilarities at PER sites were correlated exclusively with environmental distances, indicating that environmental filtering was the primary community assembly process, regardless of the inter-site distances (Brown & Swan, 2010; Heino *et al.*, 2012).

These metacommunity patterns contrast with the paradigm that headwater communities should be driven primarily by local community processes (environmental filtering), due to their spatial isolation in dendritic

Table 3 Additive partitioning of taxonomic richness for all taxa, rare and dominant taxa reported in terms of mean number of taxa (S) and percentage (%). Taxa whose relative abundance was <0.1% were considered rare; those with relative abundance >5% were considered dominant; +, – and = indicate if the number of taxa is higher, lower or not different from what would be expected by chance (see *Data analysis* for details)

	α_{sample}	β_{sample}	$\beta_{\text{site type}}$	$\gamma_{\text{Chipiriri}}$
Invertebrates				
All taxa				
S	28.1	14.5	2.5	45.0
%	62.4 (+)	32.2 (–)	5.5 (–)	100.0
Rare taxa (<0.1%)				
S	3.5	6.8	1.7	12.0
%	29.2 (+)	56.7 (=)	14.2 (–)	100.0
Dominant taxa (>1%)				
S	30.4	0.3	0	32.0
%	95.0 (+)	1.0 (–)	0 (=)	100.0
Fish				
All species				
S	43.3	19.8	4.8	68.0
%	63.7 (+)	29.1 (–)	7.1 (–)	100.0
Rare species (<0.1%)				
S	13.8	18.1	5.1	36.0
%	38.3 (+)	50.3 (=)	14.1 (–)	100.0
Dominant species (>1%)				
S	19.9	0.1	0	20.0
%	99.5 (+)	0.5 (–)	0 (=)	100.0

networks (Brown & Swan, 2010). Although this prediction has received support in perennial headwaters (Brown & Swan, 2010; Heino *et al.*, 2012; Göthe *et al.*, 2013), the results of this and recent studies (Warfe *et al.*, 2013; Cañedo-Argüelles *et al.*, 2015) suggest the role of dispersal may be underestimated in headwaters that experience disturbances such as drying. Comparing metacommunity patterns across headwater systems prone to different drying severity would be necessary to address this hypothesis.

While it is often assumed that disturbed ecosystems have communities with low α - but high β -diversity due to multiple recovery stages and/or random changes in community size (Larned *et al.*, 2010; Martin & Ferrer, 2015; Myers *et al.*, 2015), this was not totally true in the Chipiriri headwaters. Indeed, INT sites showed higher β -diversity, at least for the turnover component for invertebrates and the total β -diversity of fish communities, compared to PER sites, corroborating the contrasting local diversity and metacommunity patterns described above. The lack of difference in total invertebrate β -diversity between INT and PER sites indicated the expected higher β -diversity in ecosystems prone to disturbance due to the coexistence of multiple recovery stages and trajectories and altered community size (Larned *et al.*, 2010; Martin &

Ferrer, 2015; Myers *et al.*, 2015) may only occur temporarily, during the recovery phases. After all species have reached disturbed sites from refuges, differences in local community composition level off; community size increases and β -diversity decreases. This is probably what occurred for invertebrate communities, while fish communities were still recolonising upstream reaches during the sampling period. Nevertheless, the high river drainage diversity was essentially (>60%) due to a high α -diversity, regardless of the taxon considered. Whereas β -diversity at the sample level (i.e. among-site differences in community composition) was high when rare invertebrate and fish taxa were considered, β -diversity due to site type (i.e. differences in community composition between INT and PER sites) was always low. This is because of the broad taxonomic overlap between sites, especially for invertebrates, while nestedness, rather than species turnover, characterised fish metacommunities. Due to their high spatio-temporal dynamism (e.g. Datry *et al.*, 2015) and the multiple species encompassing a wide range of dispersal abilities, IRES may offer ideal arenas to further develop conceptual and empirical metacommunity ecology for all disturbance-prone ecosystems (Datry *et al.*, 2016).

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Supporting Information

Additional Supporting Information may be found in the online version of this article:

Appendix S1. NMDS ordinations of sites based on local invertebrate (a) and fish (b) communities using presence–absence data. White circles: INT sites ($n = 14$); black circles: PER sites ($n = 22$).

Appendix S2. Results of Mantel tests and their partial derivatives for presence–absence data based on Jaccard (Podani family) dissimilarity matrices. (a) correlations between community dissimilarities, environmental (ENV) and physical (LAND = overland, RIVER = watercourse) distances; (b) correlations between community dissimilarities and physical distances (LAND = overland, RIVER = watercourse) controlling for environmental distances (ENV); (c) correlations between community dissimilarities and environmental distances (ENV) controlling for physical distances (LAND = overland, RIVER = watercourse). Values are Mantel or partial Mantel R and associated P -values: NS $P > 0.05$, $*P < 0.05$, $**P < 0.01$, $***P < 0.001$.

Appendix S3. Invertebrate taxa collected in the Chipiriri River basin. Min, minimum; Max, maximum; SD, standard deviation. Values are numbers m^{-2} .

Appendix S4. Fish species collected in the Chipiriri River basin. Min, minimum; Max, maximum; SD, standard deviation. Values are density $site^{-1}$.

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